

Interpreting Correlation and Least-Squares Regression

Answer Key

Precalculus / Statistics

Quick Answers

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|-------------------------------------|--|
| 1. 74 | 6. (a) the value of r squared = 0.9459, — |
| 2. 71 | 7. (a) the predicted score at zero hours = 50, — |
| 3. (a) the predicted change = 12, — | 8. (a) the probability = $\frac{1}{3}$, — |
| 4. -3 | 9. (a) the predicted score = 98, — |
| 5. 0.9726 | 10. (a) the residual = -12, — |
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What is verified

10 of 16 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 3, 6, 7, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus / Statistics

HSS-ID.A – problems 1, 8

HSS-ID.B – problems 2, 4, 9, 10

HSS-ID.C – problems 3, 5, 6, 7, 8

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

4. *If they got 3: computed predicted minus actual, reversing the sign of the residual*

The data. Study time x (hours): 1, 2, 3, 4, 5, 6; test score y (points): 59, 59, 68, 74, 77, 89.
Least-squares line $\hat{y} = 50 + 6x$; $\sum(x - \bar{x})(y - \bar{y}) = 105$, $\sum(x - \bar{x})^2 = 17.5$, $\sum(y - \bar{y})^2 = 666$.

1. What score did the student who studied 4 hours earn?

Solution: Read the table under $x = 4$.

74

2. Predict the score for 3.5 hours of study.

Solution: $\hat{y} = 50 + 6(3.5) = 50 + 21$. (This is also the mean score: a least-squares line always passes through $(\bar{x}, \bar{y})$, and here $\bar{x} = 3.5$.)

$\hat{y} = 71$

3. Slope = 6. (a) predicted change over 2 extra hours. (b) interpret the slope.

Solution (a): The slope is a rate, so the predicted change is 6×2 .

12

Solution (b) — instructor-judged. Each additional hour of study is associated with a predicted increase of 6 points on the test. Full credit requires the units of both variables (hours in, points out) and language of association or prediction — not a claim that any particular student will gain exactly 6 points.

4. The 5-hour student scored 77. Find the residual.

Solution: $\hat{y} = 50 + 6(5) = 80$, so the residual is $y - \hat{y} = 77 - 80$. The negative sign says the point lies below the line.

-3

5. Compute the correlation coefficient r .

Solution:

$$r = \frac{105}{\sqrt{17.5 \times 666}} = \frac{105}{\sqrt{11655}} = \frac{105}{107.9583 \dots}$$

$r \approx 0.9726$

6. (a) Compute r^2 . (b) Interpret it.

Solution (a): $r^2 = \frac{105^2}{17.5 \times 666} = \frac{11025}{11655}$.

$r^2 \approx 0.9459$

Solution (b) — instructor-judged. About 94.6% of the variation in the test scores of these six students is accounted for by the linear relationship with study time; the remaining 5% or so is due to everything else. Full credit requires the word variation (or variability), a reference to the scores, and a percentage — restating "0.9459" is not an interpretation.

7. Intercept = 50. (a) predicted score at 0 hours. (b) is it trustworthy?

Solution (a): $\hat{y} = 50 + 6(0)$.

50

Solution (b) — instructor-judged. No. The observed study times run from 1 to 6 hours, so $x = 0$ lies outside the data. The intercept is the arithmetic value of the line there, not evidence about students who did not study at all — using it that way is extrapolation. Full credit requires naming the observed range and the word extrapolation (or an equivalent statement).

8. (a) P (a randomly chosen one of the six studied more than 4 hours). (b) why correlation is not causation.

Solution (a): Two of the six studied more than 4 hours (5 and 6), so $P = \frac{2}{6}$.

$P = \frac{1}{3}$

Solution (b) — instructor-judged. $r \approx 0.97$ says only that the six points lie close to a straight line. These are observational data: a lurking variable — prior preparation, sleep, motivation — could raise both the study time and the score, and the students chose their own study times rather than being assigned them. Only a randomized controlled experiment could support a causal claim. Full credit requires naming a plausible lurking variable or the observational design.

9. (a) Predicted score at $x = 8$. (b) why treat it with caution?

Solution (a): $\hat{y} = 50 + 6(8) = 50 + 48$.

98

Solution (b) — instructor-judged. 8 hours is beyond the largest observed study time of 6 hours, so this is an extrapolation: nothing in the data says the linear pattern continues out there. Pushed a little further the line predicts scores above 100, which the test cannot produce — a clear sign the model has left its valid range. Full credit requires the extrapolation point; the impossible-score observation is a strong bonus.

10. **Synthesis.** A seventh student studies 4 hours and scores 62. (a) residual. (b) what it means and its effect.

Solution (a): $\hat{y} = 50 + 6(4) = 74$, so the residual is $62 - 74$.

-12

Solution (b) — instructor-judged. The new student scored 12 points below what the line predicts — four times the largest residual among the original six, which was 3. A point that far off the line increases the scatter about it, so r (about 0.97) would fall, r^2 would fall with it, and the refitted line would be pulled downward in the middle of the x range, flattening the slope slightly. Full credit requires reading the sign and size of the residual and predicting a lower correlation.